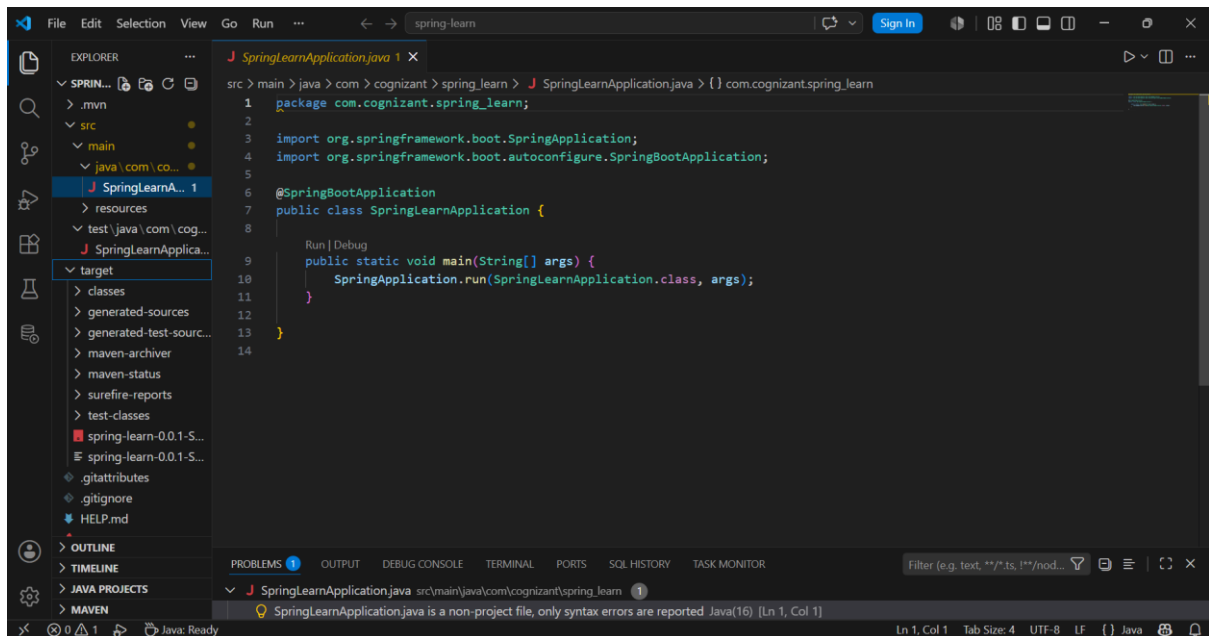
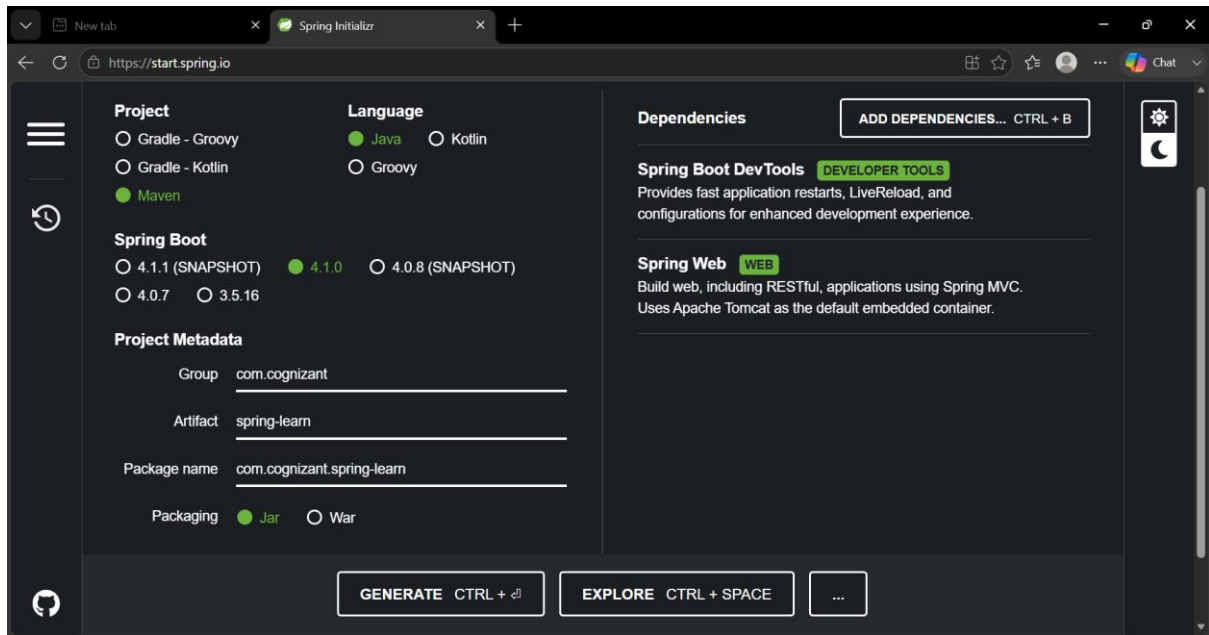


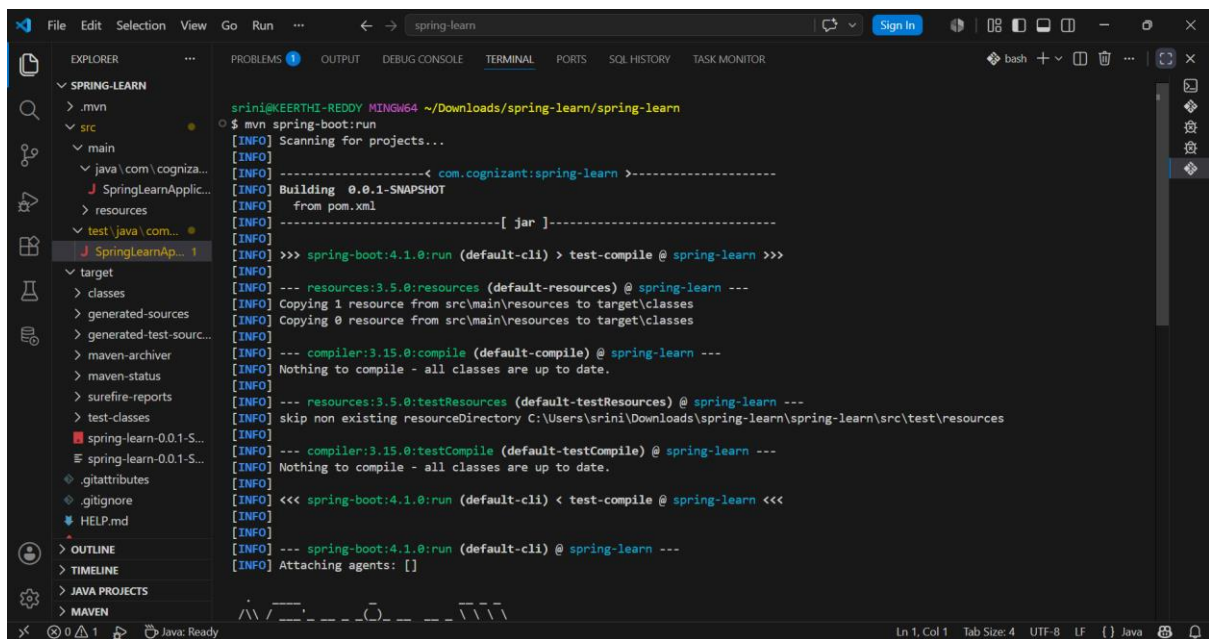
DIGITAL NURTURE 5.0 DEEPSKILLING

WEEK 3 – HANDS-ON

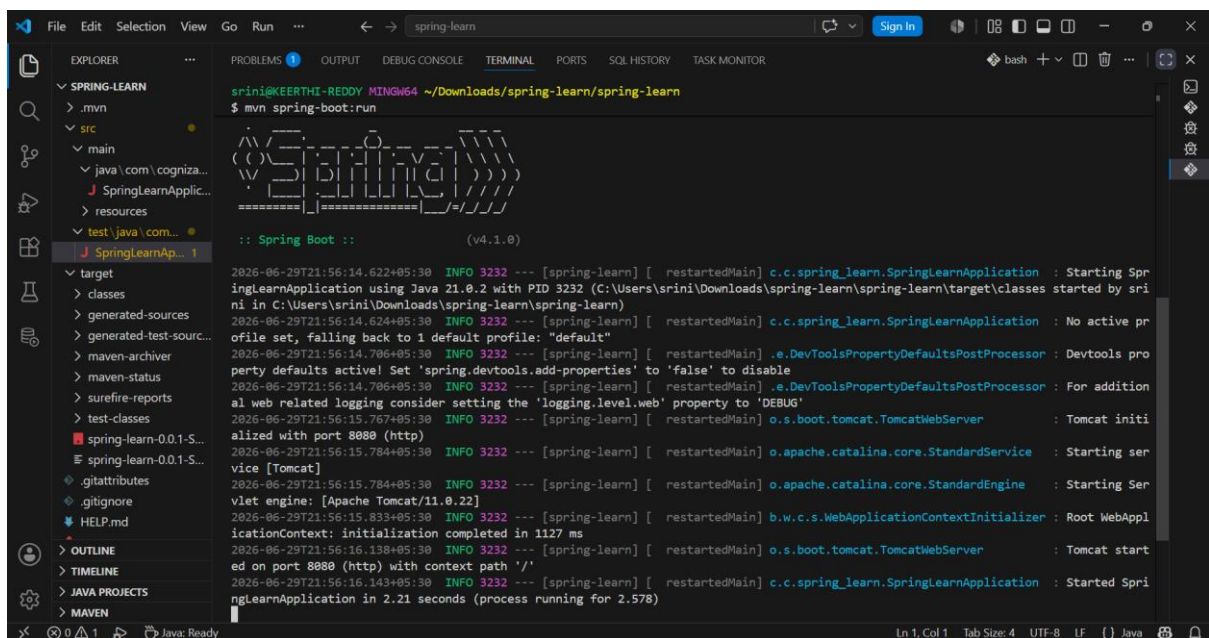
Spring REST using Spring Boot 3

Create a Spring Web Project using Maven





```
sprini@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:spring-learn >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ spring-learn >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ spring-learn ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ spring-learn ---
[INFO] skip non existing resourceDirectory C:\Users\sprini\Downloads\spring-learn\spring-learn\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ spring-learn <<<
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []
```



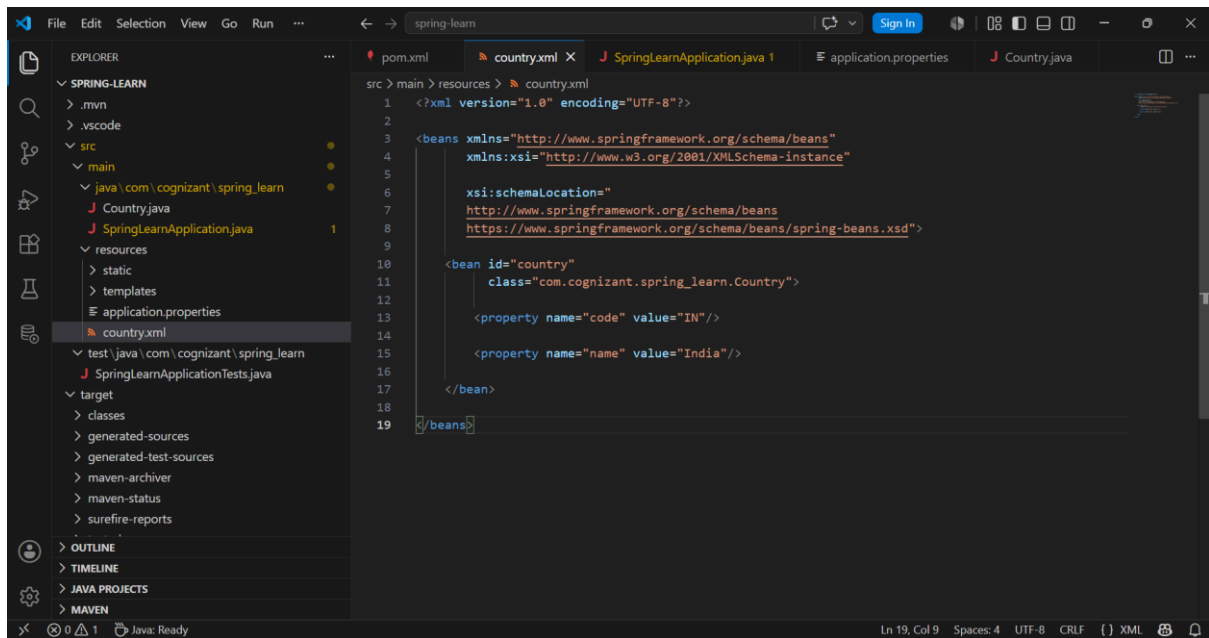
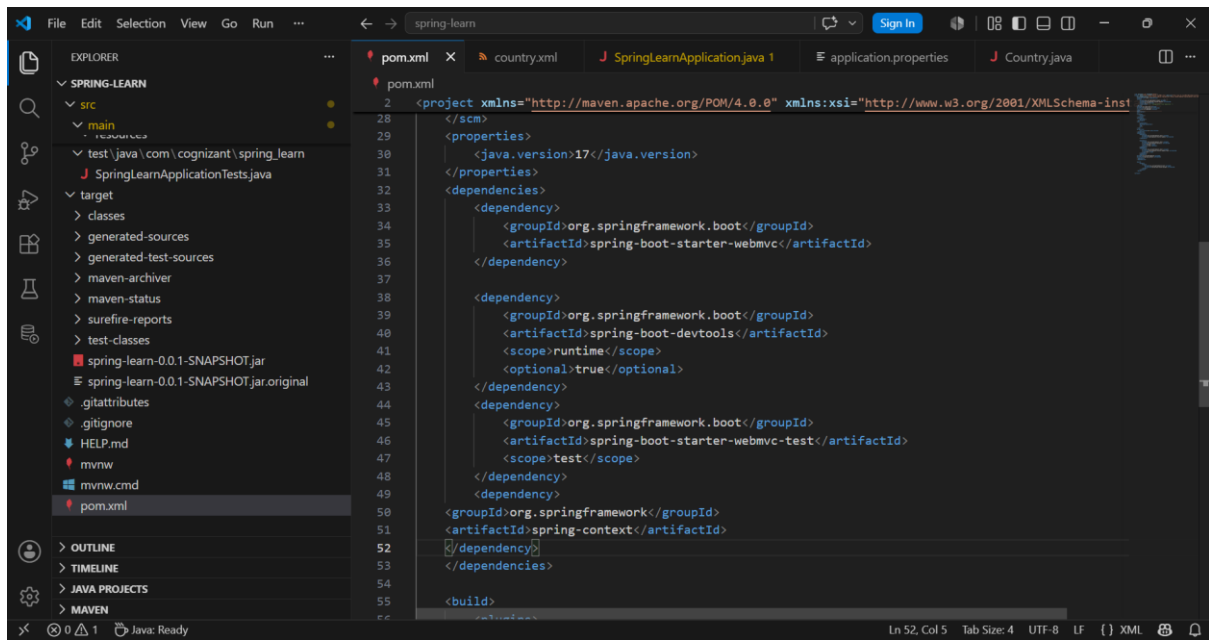
```
sprini@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run

  ____ _
 / ___ \| | | |
/ /___ \| |_| |
 \___ \|____|_|_|

:: Spring Boot ::      (v4.1.0)

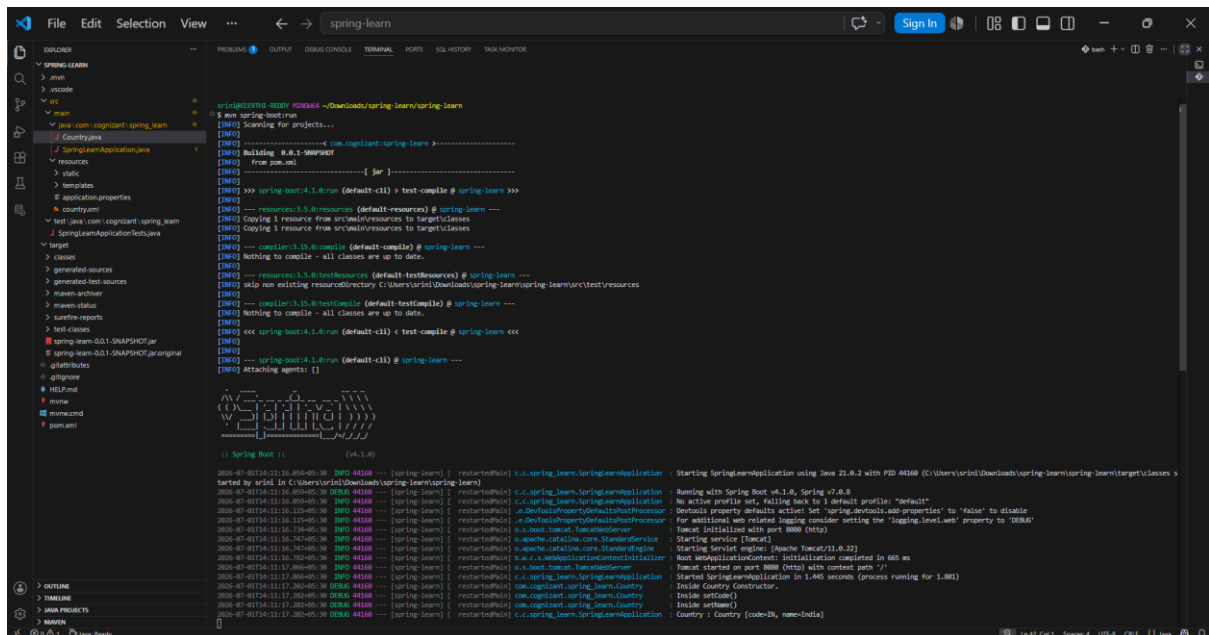
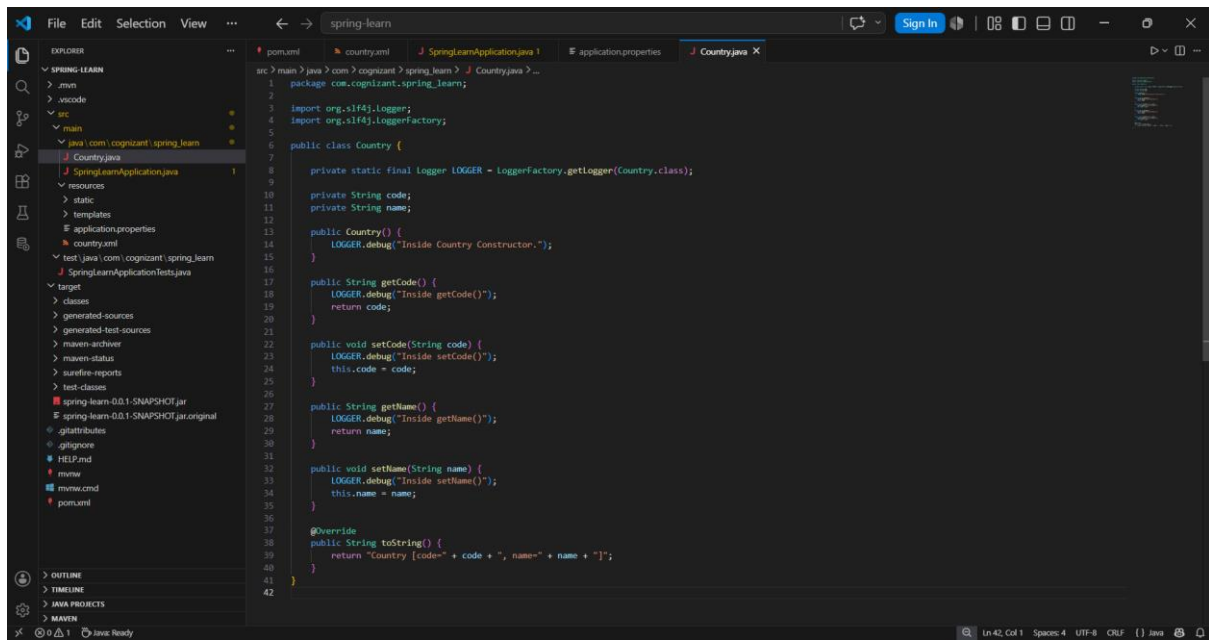
2026-06-29T21:56:14.622+05:30 INFO 3232 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting Spr
ingLearnApplication using Java 21.0.2 with PID 3232 (C:\Users\sprini\Downloads\spring-learn\spring-learn\target\classes started by sri
ni in C:\Users\sprini\Downloads\spring-learn\spring-learn)
2026-06-29T21:56:14.624+05:30 INFO 3232 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active pr
ofile set, falling back to 1 default profile: "default"
2026-06-29T21:56:14.706+05:30 INFO 3232 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools pro
perty defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-06-29T21:56:14.706+05:30 INFO 3232 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For addition
al web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-06-29T21:56:15.767+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initi
alized with port 8080 (http)
2026-06-29T21:56:15.784+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting ser
vice [Tomcat]
2026-06-29T21:56:15.784+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Ser
vlet engine: [Apache Tomcat/11.0.22]
2026-06-29T21:56:15.833+05:30 INFO 3232 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebAppl
icationContext: initialization completed in 1127 ms
2026-06-29T21:56:16.138+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat start
ed on port 8080 (http) with context path '/'
2026-06-29T21:56:16.143+05:30 INFO 3232 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started Spr
ingLearnApplication in 2.21 seconds (process running for 2.578)
```

Spring Core – Load Country from Spring Configuration XML



```
src > main > java > com > cognizant > spring_learn > J SpringLearnApplication.java > SpringLearnApplication
1 package com.cognizant.spring_learn;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8
9 import org.springframework.boot.SpringApplication;
10 import org.springframework.boot.autoconfigure.SpringBootApplication;
11
12 @SpringBootApplication
13 public class SpringLearnApplication {
14
15     // Logger declaration
16     private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class);
17
18     Run | Debug
19     public static void main(String[] args) {
20         SpringApplication.run(SpringLearnApplication.class, args);
21
22         // Call the method to load country bean
23         displayCountry();
24     }
25
26     public static void displayCountry() {
27
28         // Load the Spring XML configuration
29         ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
30
31         // Get the country bean
32         Country country = context.getBean("country", Country.class);
33
34         // Display country details
35         LOGGER.debug("Country : {}", country.toString());
36     }
37
38 }
```

```
src > main > resources > application.properties
1 spring.application.name=spring_learn
2 logging.level.com.cognizant.spring_learn=DEBUG
```



Hello World RESTful Web Service

This screenshot shows an IDE window with the file `spring-learn` open. The Explorer sidebar on the left displays the project structure, including `src/main/java/com/cognizant/spring_learn/controller`. The main editor area shows the `HelloController.java` file with the following code:

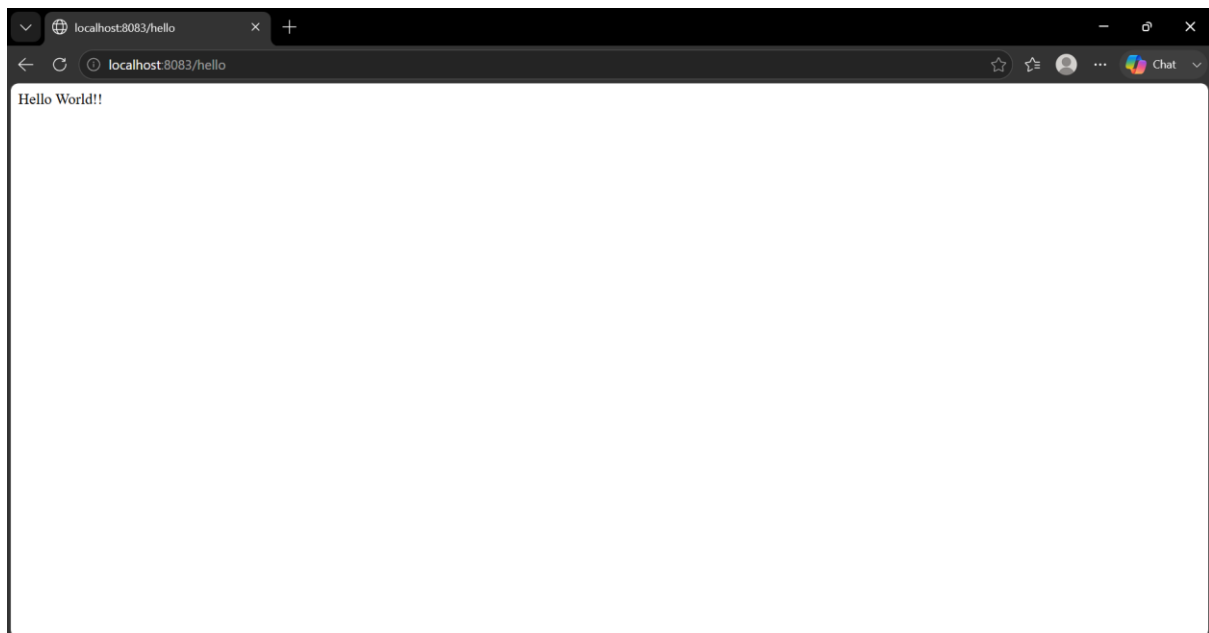
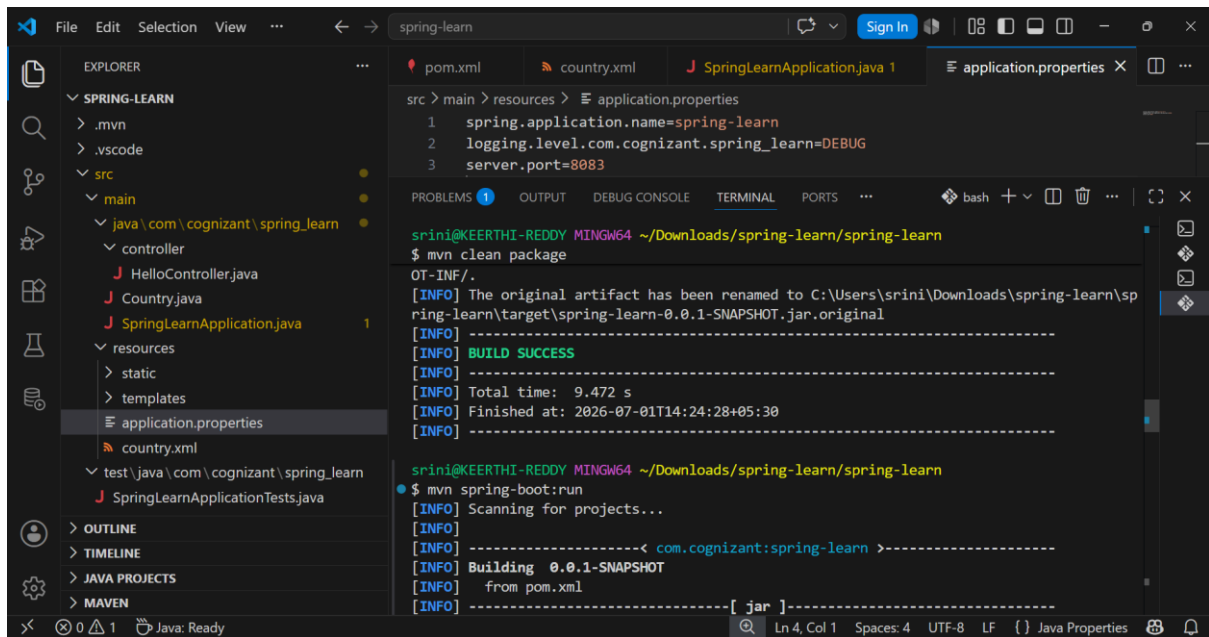
```
1 package com.cognizant.spring_learn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 import org.springframework.web.bind.annotation.GetMapping;
7 import org.springframework.web.bind.annotation.RestController;
8
9 @RestController
10 public class HelloController {
11
12     private static final Logger LOGGER =
13         LoggerFactory.getLogger(HelloController.class);
14
15     @GetMapping("/hello")
16     public String sayHello() {
17
18         LOGGER.info("START - sayHello()");
19
20         String message = "Hello World!!";
21
22         LOGGER.info("END - sayHello()");
23
24         return message;
25     }
26
27 }
28
```

The status bar at the bottom indicates the cursor is at line 28, column 1, with 4 spaces, UTF-8 encoding, and CRLF line endings.

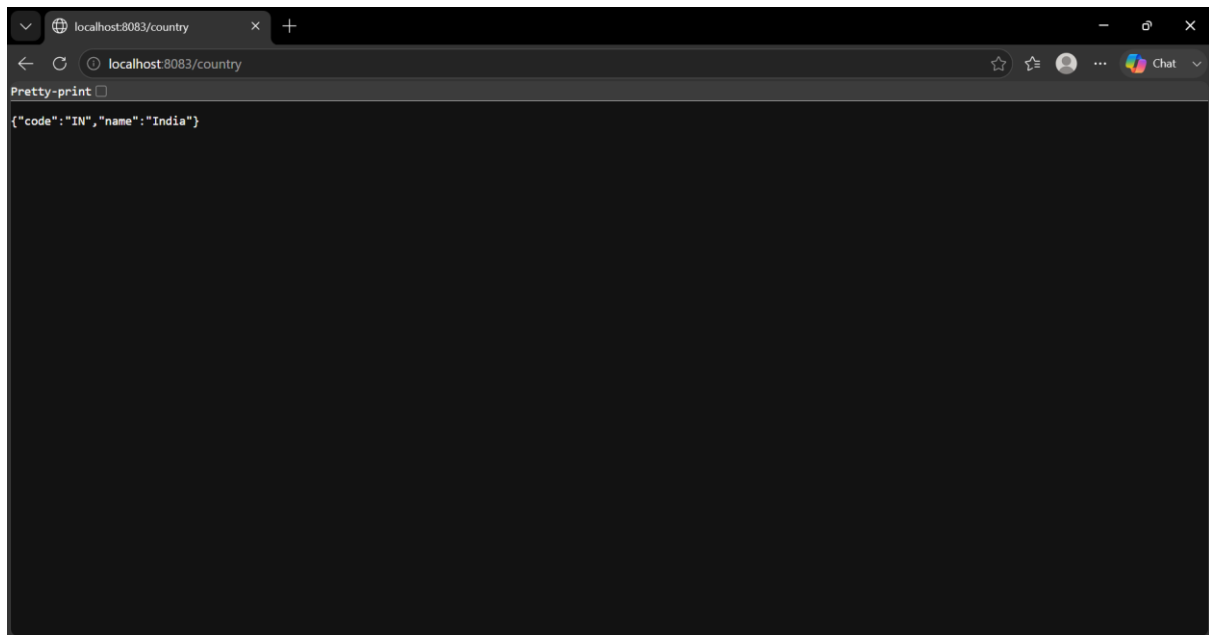
This screenshot shows the same IDE window with the `application.properties` file open in the main editor. The Explorer sidebar shows the file selected under `src/main/resources`. The content of the file is:

```
1 spring.application.name=spring-learn
2 logging.level.com.cognizant.spring_learn=DEBUG
3 server.port=8083
4
```

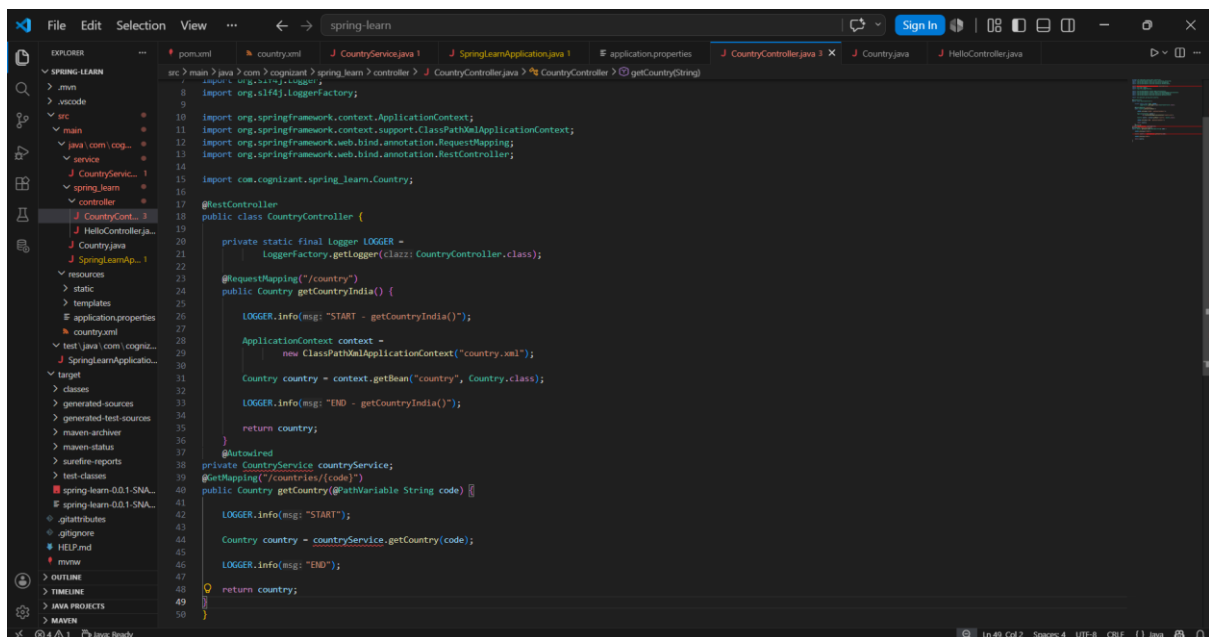
The status bar at the bottom indicates the cursor is at line 4, column 1, with 4 spaces, UTF-8 encoding, and LF line endings.



REST - Country Web Service



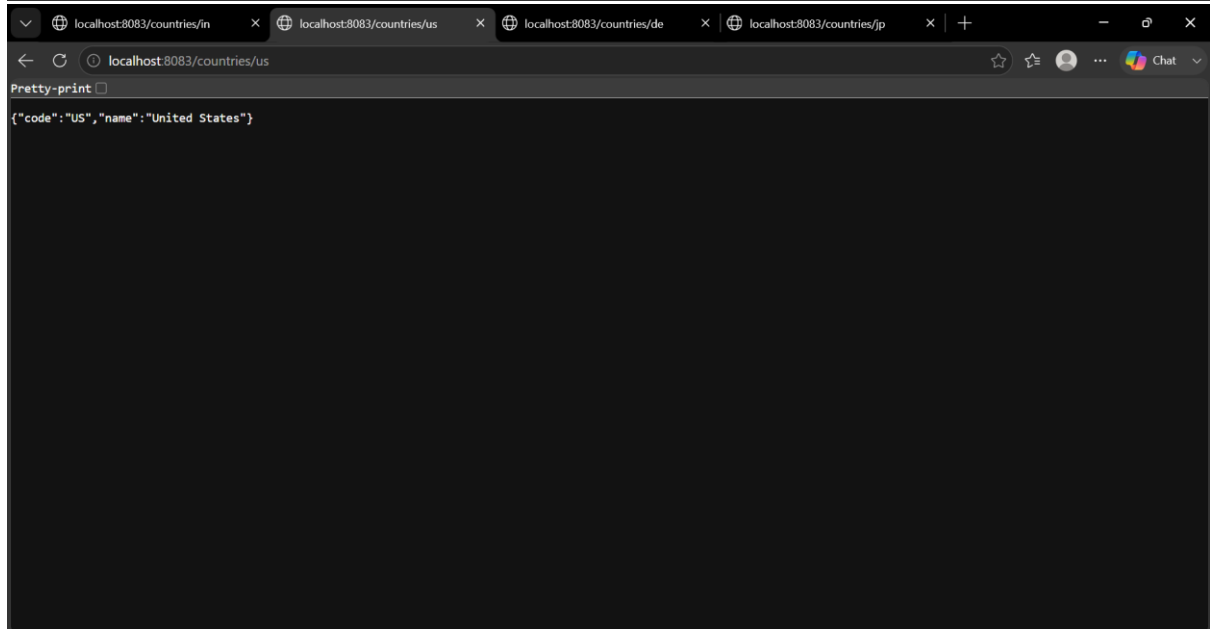
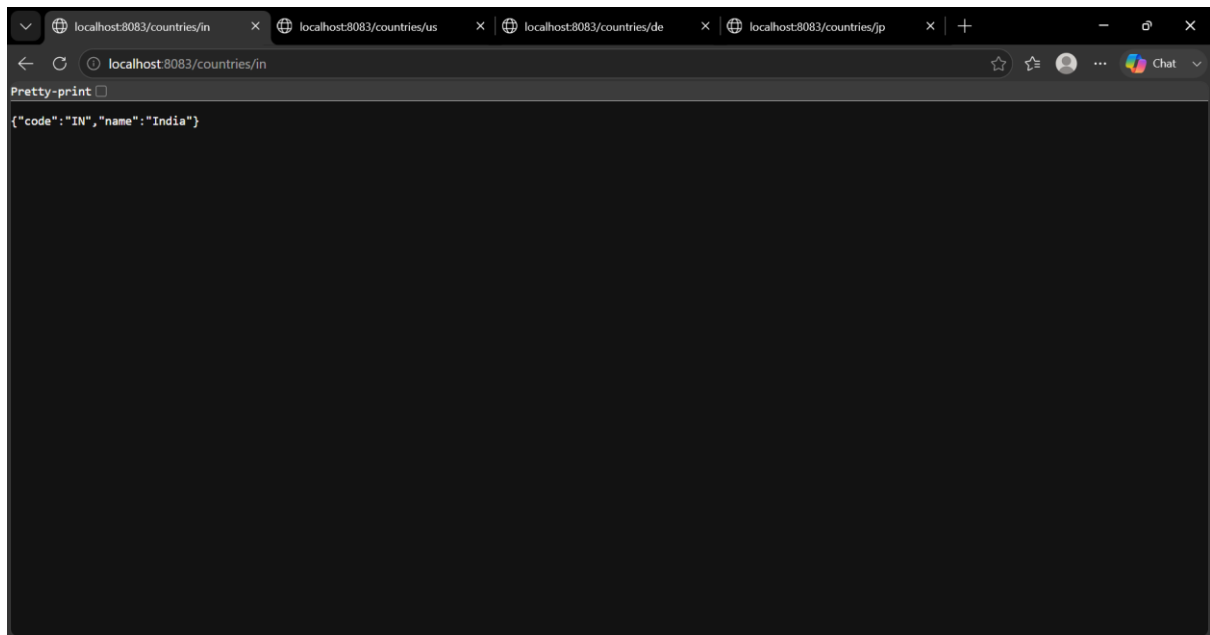
REST - Get country based on country code

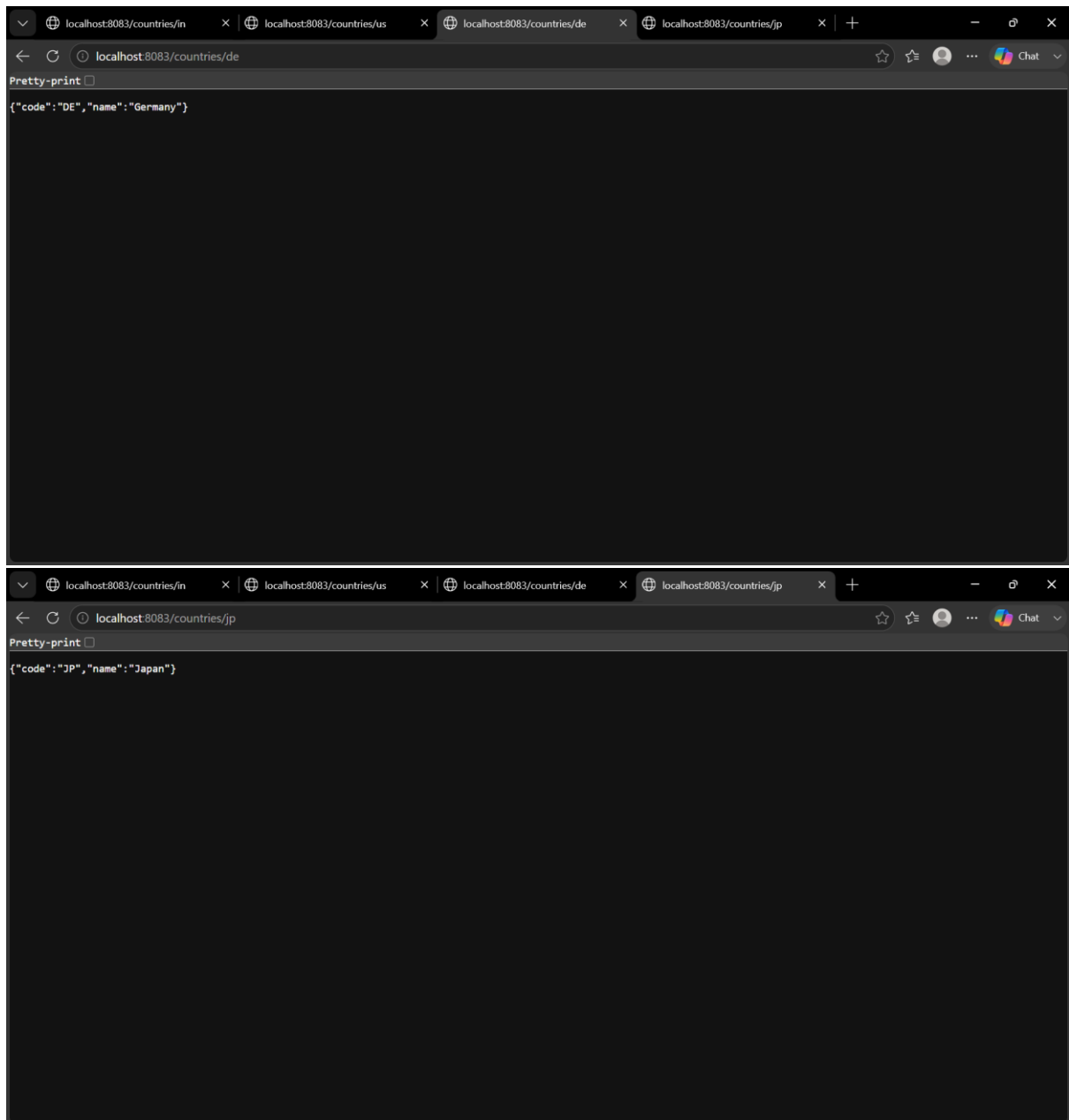


```
src > main > java > com > cognizant > service > J CountryService.java > % CountryService
1 package com.cognizant.spring_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.stereotype.Service;
8
9 import com.cognizant.spring_learn.Country;
10
11 @Service
12 public class CountryService {
13
14     @SuppressWarnings("unchecked")
15     public Country getCountry(String code) {
16
17         ApplicationContext context =
18             new ClassPathXmlApplicationContext("country.xml");
19
20         List<Country> countryList =
21             (List<Country>) context.getBean("countryList");
22
23         for (Country country : countryList) {
24
25             if (country.getCode().equalsIgnoreCase(code)) {
26                 return country;
27             }
28
29         }
30
31         return null;
32     }
33 }
```

```
src > main > java > com > cognizant > service > J CountryService.java > % CountryService
1 package com.cognizant.spring_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.stereotype.Service;
8
9 import com.cognizant.spring_learn.Country;
10
11 @Service
12 public class CountryService {
13
14     @SuppressWarnings("unchecked")
15     public Country getCountry(String code) {
16
17         ApplicationContext context =
18             new ClassPathXmlApplicationContext("country.xml");
19
20         List<Country> countryList =
21             (List<Country>) context.getBean("countryList");
22
23         for (Country country : countryList) {
24
25             if (country.getCode().equalsIgnoreCase(code)) {
26                 return country;
27             }
28
29         }
30
31         return null;
32     }
33 }
```

```
spring@KEERTHI-REDDY KING64: ~/Downloads/spring_learn/spring_learn
$ mvn spring-boot:run
2026-07-01T14:41:45.021+05:30 INFO 11804 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-01T14:41:45.108+05:30 INFO 11804 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-01T14:41:45.108+05:30 INFO 11804 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-01T14:41:45.272+05:30 INFO 11804 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-01T14:41:45.392+05:30 INFO 11804 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T14:41:45.397+05:30 INFO 11804 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-01T14:41:46.057+05:30 INFO 11804 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 948 ms
2026-07-01T14:41:46.341+05:30 WARN 11804 --- [spring-learn] [ restartedMain] ConfigServletWebServerApplicationContext : Exception encountered during context initialization - cancelling refresh attempt: org.springframework.context.ApplicationContextException: Failed to start bean 'webServerStartStop'
2026-07-01T14:41:46.349+05:30 INFO 11804 --- [spring-learn] [ restartedMain] s.b.a.l.ConditionEvaluationReportLogger :
Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.
2026-07-01T14:41:46.366+05:30 ERROR 11804 --- [spring-learn] [ restartedMain] o.s.b.d.LoggingFailureAnalysisReporter :
*****
APPLICATION FAILED TO START
*****
Description:
Web server failed to start. Port 8083 was already in use.
Action:
Identify and stop the process that's listening on port 8083 or configure this application to listen on another port.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.118 s
[INFO] Finished at: 2026-07-01T14:41:46+05:30
[INFO] -----
spring@KEERTHI-REDDY KING64: ~/Downloads/spring_learn/spring_learn
$
```





Create authentication service that returns JWT

